Comments on TN266v1.4 by Megan

–Replies (http://www.t2k.org/docs/technotes/266/Comments/reply_Megan_v1p0.pdf), line 3-30 → Did you add this info to the note somewhere? Can you add it? (I couldn't find it..) I think it's useful.

DONE. Appendix E is added. Also caption of tables 18 - 21 has been improved.

– Replies, line 45-48 → Again, can you add this to the note? I couldn't find it.

Added in appendix E.

– Replies, line 128-129 → Does starting at the boundary of the fit range really remove possible local minima problems in the fit? I would think that would actually make it worse, since the fitter can only start stepping in 1 direction... anyway it may be worth it to think about expanding the fitter range to $\pm 2\pi$ in future fits.

Since all the nuisance parameters are marginalized, i.e. the fit parameters are not floating in n-dimensional space, using the method shown in section 5.4 the global minimum is found almost 100% of the times. However, when the number of free parameters minimized is increased (more than 2), fit strategy should be revisited and optimized.

– Replies, line 135-137 → Although the IH is excluded at $> 90\%$ CL, T2K doesn't actually have the sensitivity to actually exclude the IH (as was discussed at length last week). Then I think it's important to include the IH-only limits in the note as well.

DONE.

– TN266v1.4, line 780 + 783 + fig 20 : description doesn't match figure caption (IH vs NH).

DONE. Caption of fig 20 was wrong. Best-fit is inverted hierarchy, not normal.

– TN266v1.4, line 875-877, 878-879 : I'm suddenly worried about fitter biases pointing our data toward Asimov Set A. Line 878-879 – this may be true now (and I'm not suggesting delaying the release of this analysis), but since these things don't agree within error bars for all parameter values, may need to re-think this for later versions of this analysis. (I guess it's worse if things far away from our best fit point don't make sense, right?)..

The test values of $\sin^2\theta_{23}$ are quite extreme, indeed when performing the posterior predictive method the critical values are close enough. Also the difference when $\sin^2\theta_{23}$ is fixed is not very large, though not fully negligible. I think it's safe for this analysis release. However we need to make more studies for when higher statistic is collected.

– Table 27 caption : Asimov set A has $\delta_{CP} = -\pi/2$, right?

Yes it's correct. So the fake data sets were built by marginalizing all the oscillation parameters and fixing $\delta_{CP} = 0$ and normal hierarchy. Then the fake data sets were weighted using the Asimov data set A, assuming it is the data set I'm fitting.